INTERNAL HOM FROM A FINITELY PRESENTED MODULE INTO A QC MODULE IS LQC-FBC AND HAS THE TENSOR ADJOINT

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Internal hom sheafifies. Modules on Sites, Lemma sites-modules-lemma-internal-hom. fppf sheaf property.
- fbc of Hom from fp modules. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-check-flat-comparison-on-etale-covering (6). Flat base change property.
- Internal hom tensor adjunction. Modules on Sites, Lemma sites-modules-lemma-internal-hom-adjoint-tensor. Final universal property.

Source reference. Original reference 04TT: sources/stacks-chow/tags.txt, source lines 6266–6266. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03EM: sources/stacks-chow/tags.txt, source lines 4461–4461. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06WM: sources/stacks-chow/tags.txt, source lines 8821–8821. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0764: sources/stacks-chow/tags.txt, source lines 9154–9154. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0778: sources/stacks-chow/tags.txt, source lines 9193–9193. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03EO: sources/stacks-chow/tags.txt, source lines 4463–4463. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. Let $\mathcal{F}$ be an $\mathcal{O}_\mathcal{X}$-module of finite presentation and let $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_\mathcal{X}$-module. The internal homs $\mathcal{H}om_{\mathcal{O}_\mathcal{X}}(\mathcal{F}, \mathcal{G})$ computed in $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$ or $Mod(\mathcal{O}_\mathcal{X})$ agree and the common value is an object*

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of $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$. The quasi-coherent module $hom(\mathcal{F},\mathcal{G}) = Q(\mathcal{H}om_{\mathcal{O}_\mathcal{X}}(\mathcal{F},\mathcal{G}))$ has the following universal property

$$\mathrm{Hom}_\mathcal{X}(\mathcal{H}, hom(\mathcal{F},\mathcal{G})) = \mathrm{Hom}_\mathcal{X}(\mathcal{H} \otimes_{\mathcal{O}_\mathcal{X}} \mathcal{F}, \mathcal{G})$$

for $\mathcal{H}$ in $QCoh(\mathcal{O}_\mathcal{X})$.

Proof. The construction of $\mathcal{H}om_{\mathcal{O}_\mathcal{X}}(\mathcal{F},\mathcal{G})$ in Modules on Sites, Section 04TT depends only on $\mathcal{F}$ and $\mathcal{G}$ as presheaves of modules; the output $\mathcal{H}om$ is a sheaf for the fppf topology because $\mathcal{F}$ and $\mathcal{G}$ are assumed sheaves in the fppf topology, see Modules on Sites, Lemma 03EM. By Sheaves on Stacks, Lemma 06WM we see that $\mathcal{H}om_{\mathcal{O}_\mathcal{X}}(\mathcal{F},\mathcal{G})$ is locally quasi-coherent. By Lemma 0764 we see that $\mathcal{H}om_{\mathcal{O}_\mathcal{X}}(\mathcal{F},\mathcal{G})$ has the flat base change property. Hence $\mathcal{H}om_{\mathcal{O}_\mathcal{X}}(\mathcal{F},\mathcal{G})$ is an object of $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$ and it makes sense to apply the functor Q of Lemma 0778 to it. By the universal property of Q we have

$$\mathrm{Hom}_\mathcal{X}(\mathcal{H}, Q(\mathcal{H}om_{\mathcal{O}_\mathcal{X}}(\mathcal{F},\mathcal{G}))) = \mathrm{Hom}_\mathcal{X}(\mathcal{H}, \mathcal{H}om_{\mathcal{O}_\mathcal{X}}(\mathcal{F},\mathcal{G}))$$

for $\mathcal{H}$ quasi-coherent, hence the displayed formula of the lemma follows from Modules on Sites, Lemma 03EO. □